Structural Engineering Principles

Technical Reference for Load Analysis, Foundation Design, and Material Behavior

1. Load Types and Structural Analysis

Structural engineering begins with the accurate identification and quantification of all loads acting on a structure. Building codes categorize loads into two primary groups: gravity loads (dead and live) and lateral loads (wind and seismic). Dead loads are permanent forces resulting from the self-weight of all structural components (beams, columns, slabs, walls) and non-structural components (finishes, mechanical equipment, ceiling systems). Live loads are variable, temporary forces produced by building occupancy, furniture, vehicles, and other movable loads. Design codes specify minimum live load requirements based on occupancy classification, but actual live loads should be verified against the intended use of the structure.

Wind loads are determined through a systematic process that considers basic wind speed at the building location (obtained from wind speed maps), terrain exposure categories (B, C, or D reflecting surface roughness), building height and geometry, and the importance factor based on the building's risk category. The ASCE 7 standard provides three methods for wind load calculation: the simplified method (applicable to simple diaphragm buildings), the analytical method (envelope procedure for low-rise buildings and directional procedure for all buildings), and the wind tunnel method (required for complex, flexible, or dynamically sensitive structures). Each method produces design pressures for main wind force resisting systems (MWFRS) and components and cladding (C&C;), with C&C; pressures typically being significantly higher due to localized wind effects at corners, ridges, and eaves.

Load Type	Description	Typical Values / Ranges	Governing Standard
Dead Load (DL)	Self-weight of structural and non-structural elements	Floor: 2.0-7.0 kN/m2; Roof: 1.0-3.5 kN/m2	ASCE 7, EN 1991
Live Load (LL)	Occupancy, furniture, movable equipment	Office: 2.4-3.6 kN/m2; Storage: 6.0-12.0 kN/m2	ASCE 7 Table 4-1
Wind Load	Static and dynamic pressure from wind	Varies by region, height, exposure (20-5.0 kN/m2)	ASCE 7 Ch.26-31
Seismic Load	Inertial forces from ground acceleration	Depends on Ss, S1, Site Class, R factor	ASCE 7 Ch.11-23, IBC
Snow Load	Weight of accumulated snow on roof	Ground snow: 0.5-10.0 kN/m2 (by location)	ASCE 7 Ch.7
Rain Load	Accumulated rainwater on low-slope roofs	Based on drainage capacity and roof slope	ASCE 7 Ch.8

Table 1: Structural Load Types and Design Values

2. Foundation Systems

Foundation engineering involves the design of structural elements that transfer building loads to the ground while limiting settlements to acceptable levels. The choice of foundation system depends on the magnitude and distribution of applied loads, the bearing capacity and compressibility of underlying soils, groundwater conditions, site constraints, and economic considerations. Geotechnical investigation, including borings, laboratory testing, and in-situ tests (SPT, CPT), provides the essential soil parameters needed for foundation design: bearing capacity, shear strength, consolidation characteristics, and groundwater level.

Shallow foundations (spread footings, strip footings, and mat foundations) are the most economical option when competent bearing soil is available at relatively shallow depths (typically less than 3 meters). The bearing capacity of shallow foundations is calculated using Terzaghi's or Meyerhof's equations, which consider soil shear strength, foundation geometry, embedment depth, and load inclination. Mat foundations distribute column loads over the entire building footprint, reducing bearing pressures and providing additional rigidity that helps limit differential settlement. When shallow foundations would result in excessive settlement or bearing capacity failure, deep foundations (piles or drilled shafts) transfer loads to deeper, more competent soil or rock strata through end bearing, skin friction, or a combination of both.

Foundation Type	Bearing Capacity (kPa)	Depth Range	Best Soil Conditions	Typical Applications
Spread Footing	100-400+	0.5-2.0 m	Stiff clay, dense sand, rock	Individual columns, light structures
Strip Footing	100-300	0.5-1.5 m	Firm to stiff soils	Load-bearing walls, residential
Mat (Raft) Foundation	50-200	1.0-3.0 m	Soft to medium soils, variable fill	Heavy buildings, uniform loads
Driven Pile	200-2000+	10-30+ m	Soft soils over firm strata	High-rises, waterfront, bridges
Drilled Shaft (Caisson)	500-5000+	10-40+ m	Stiff clay, rock socket	Bridges, heavy industrial
Micropile	200-1000	5-20 m	Restricted access, poor soil	Underpinning, retrofit, landslide

Table 2: Foundation Types and Applicability

3. Structural Material Properties

Understanding the mechanical properties of construction materials is fundamental to structural design. The elastic modulus (Young's modulus) defines the stiffness of a material and determines how much it deforms under load, while the yield strength (for ductile materials) or compressive strength (for brittle materials) defines the stress at which permanent deformation or failure begins. The ratio of yield strength to density, known as specific strength, is particularly important in design

because it indicates how efficiently a material carries load relative to its self-weight, which is a critical consideration for long-span and tall structures.

Concrete exhibits non-linear stress-strain behavior, with a relatively low tensile strength (approximately 8-12% of its compressive strength) that necessitates the use of steel reinforcement to resist tension, shear, and sometimes compression in structural members. The Eurocode 2 and ACI 318 design codes use different but equivalent approaches for concrete member design, with ACI 318 employing strength design (load and resistance factor design, LRFD) that applies strength reduction factors to nominal member capacities and load factors to applied loads to achieve target reliability levels. Steel, by contrast, behaves as a linear-elastic material up to the yield point, after which it undergoes significant plastic deformation (strain hardening) before fracture, providing valuable ductility and energy absorption capacity that is essential for seismic design.

Material	Elastic Modulus (GPa)	Yield Strength (MPa)	Density (kg/m3)	Thermal Expansion (x10-6/C)
Structural Steel (A992)	200	345-450	7850	12.0
Concrete (C30/37)	25-33	30 (compressive)	2400	10.0
Reinforcing Steel (Grade 60)	200	420	7850	12.0
Douglas Fir-Larch (Select)	12-13	7.6 (bending)	480-510	3.5-4.5
Aluminum 6061-T6	68.9	276	2700	23.6
Masonry (CMU, grouted)	10-20	10-25 (compressive)	1900-2100	5.0-8.0

Table 3: Mechanical Properties of Common Structural Materials

4. Design Codes and Standards

Structural design in the United States is governed by a hierarchy of codes and standards. The International Building Code (IBC) establishes the overarching requirements for structural design, referencing specific material standards for detailed design procedures: ACI 318 for reinforced concrete structures, AISC 360 for steel structures, NDS for wood structures, and TMS 402/602 for masonry structures. These material standards are complemented by ASCE 7 (Minimum Design Loads and Associated Criteria for Buildings and Other Structures), which defines the load combinations, load factors, and environmental load calculation procedures that all material-specific standards must follow.

The transition from allowable stress design (ASD) to load and resistance factor design (LRFD) represents one of the most significant advances in structural engineering practice over the past half-century. LRFD provides a more rational and consistent level of reliability across different loading conditions and material types by applying separate factors to loads (increasing them to account for uncertainty) and resistances (reducing them to account for material variability and

model uncertainty). The fundamental LRFD equation requires that the factored resistance (phi times Rn) must equal or exceed the factored load effect (sum of gamma times Q), where phi is the resistance factor, Rn is the nominal strength, gamma is the load factor, and Q represents the load effect.

International design codes follow similar principles with regional adaptations. The Eurocodes (EN 1990 through EN 1999) provide a comprehensive set of harmonized technical rules for the structural design of construction works across the European Union, using partial safety factors for both actions (loads) and material properties. Similarly, countries such as Canada (CSA standards), Australia (AS/NZS standards), and Japan (JSCE standards) maintain their own code systems that reflect local material availability, construction practices, seismicity, and climate conditions. Structural engineers working on international projects must be familiar with the applicable local codes and any required deviations from their domestic practice, particularly regarding load combinations, seismic design categories, and material testing standards.